

Hexian Wang

Meteorological Foundation Model Algorithm Engineer | Weather and Power Forecasting Researcher

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🎓 Google Scholar | 🌐 GitHub | 🆔 ORCID

Research Interests

Weather foundation models; renewable-energy power forecasting; AI agents for energy systems; time-series foundation models; probabilistic forecasting; large-scale AI systems for weather and energy.

Education

Shanxi University

Sep. 2023 – Jul. 2026

M.Sc. in Control Science and Engineering; GPA: 3.50/4.00

Taiyuan, China

- Research focus: wind power forecasting, time-series foundation models, and large-scale model training.

The University of Western Australia

Nov. 2024 – May 2025

Government-Sponsored Visiting Research Student

Perth, Australia

- Host supervisor: [Professor Tongming Zhou](#); conducted research on AI-driven wind power forecasting and large-scale experiments.

Henan Institute of Science and Technology

Sep. 2019 – Jun. 2023

B.Eng. in Electrical Engineering and Automation; GPA: 3.48/4.00

Xinxiang, China

Industry Experience

Ming Yang Smart Energy Group Beijing Technology Co., Ltd.

Jul. 2026 – Present

Meteorological Foundation Model Algorithm Engineer

Beijing, China

Subsidiary of Ming Yang Smart Energy Group (SSE: 601615), a global wind-energy technology company.

- Develop weather forecasting foundation models for renewable-energy applications using large-scale numerical weather prediction, reanalysis, and observational data.
- Build data-processing, training, inference, evaluation, and deployment workflows informed by state-of-the-art weather models, including Pangu-Weather, GraphCast, and Aurora.
- Adapt meteorological foundation-model outputs to downstream wind and solar power forecasting, connecting advanced weather AI with operational renewable-energy decision support.

Publications

[1] [Wang, H.](#), Guo, D., Wang, L., Zhou, T., Jia, C., and Liu, Y. “A novel frequency sparse downsampling interaction transformer for wind power forecasting.” *Energy*, vol. 326, 136199, 2025. **First author; Journal Impact Factor: 10.1.**
🌐 github.com/Hexian-2001/FSDIformer

[2] [Wang, H.](#), Zhou, T., Jia, C., Liu, Y., and Wang, L. “DeepWind: A foundation model for zero-shot wind power forecasting.” *Energy*, vol. 360, 141793, 2026. **First author; Journal Impact Factor: 10.1.**
🌐 github.com/Hexian-2001/DeepWind | 🤗 [Hugging Face: DeepWind1.0-890M](#)

Selected Research Experience

DeepWind: A Domain Foundation Model for Wind Power Forecasting

Sep. 2025 – May 2026

First author and lead developer

- Developed a decoder-only Transformer with a Regime-Aware Mixture-of-Experts architecture for zero-shot and few-shot probabilistic wind power forecasting across unseen sites.
- Curated approximately 560 billion multi-source time-series data points and designed WindBench, an eight-dataset benchmark with strict separation between pretraining and evaluation data.
- Trained models of up to 1.3B parameters on 64 AMD MI250X GPUs across eight Pawsey supercomputing nodes using distributed data parallelism, mixed precision, and gradient accumulation.
- Reduced average zero-shot nMAE by 20.86% relative to the strongest competing time-series foundation model; released source code and model weights publicly.

FSDIformer: Frequency-Aware Transformer for Wind Power Forecasting

Nov. 2024 – May 2025

First author and lead developer

- Proposed frequency-sparse attention and downsampling interaction blocks to model noisy, non-stationary wind-power series and short- and long-range temporal dependencies.

Selected Research Experience (continued)

- Benchmarked FSDIformer against Informer, Autoformer, iTransformer, TimesNet, and TimeMixer on measured wind-farm data; improved errors by 11.87% for ultra-short-term and 5.97% for short-term forecasting over the second-best model.

Intelligent Wind and Solar Power Forecasting
Team leader, State Grid competition; researcher, complex-terrain wind forecasting project

- Led modelling of 0–24 h and 1–10 day renewable-power forecasts by integrating historical generation, numerical weather prediction, atmospheric information, sensor observations, and turbine operating features.

Jun. 2024 – Oct. 2024

Research and Technical Skills

Weather AI	Weather forecasting foundation models; Pangu-Weather, GraphCast, and Aurora; meteorological data processing; model adaptation, evaluation, inference, and deployment.
Energy AI	Wind and solar power forecasting; deterministic and probabilistic prediction; zero-shot and few-shot transfer; graph neural networks for spatiotemporal modelling.
Foundation Models	Transformer and mixture-of-experts architectures, Mamba, long-sequence modelling, pretraining, fine-tuning, and post-training.
AI Agents & Coding	Agentic workflows for scientific computing and energy systems; coding-assisted model development, experiment automation, evaluation, and deployment.
Distributed Training	PyTorch, PyTorch Lightning, Hugging Face, DDP, FSDP, DeepSpeed, mixed precision, and multi-node multi-GPU training.
Data and Systems	Python, NumPy, pandas, SQL, GluonTS, Linux, Shell, SLURM, Git, HPC experiment management, and model deployment.
English	IELTS Academic Overall: 6.5; Listening: 6.0; Reading: 7.5; Writing: 6.5; Speaking: 6.0.

Honors and Awards

National Scholarship for Graduate Students, Ministry of Education	2025
Outstanding Master’s Thesis Award, Shanxi University	2026
Graduate Academic Scholarship, Shanxi University	2025
Outstanding Graduate, Henan Institute of Science and Technology	2023

Academic Referees

Professor Tongming Zhou Department of Civil, Environmental and Mining Engineering The University of Western Australia Host supervisor and research collaborator tongming.zhou@uwa.edu.au	Professor Lingmei Wang School of Automation and Software Shanxi University Master’s supervisor 19834525388@163.com
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